

"The technology being used to make our highways smarter is on a par with the best globally. The new projects must have these elements embedded in the project cost, as retrofitting can be financially complex"

– VIJAY CHHIBBER
Former Secretary, Ministry of Road Transport & Highways

inputs, triggers alarms and flashes warning messages on LED displays along the expressway.

WHY IT IS A GAME CHANGER

Smart infrastructure is the blend of technologies integrated with modern infrastructure that can be used by consumers of public utilities to make efficient and quick decisions. Smart highways are such stretches where digital sensors are deployed to acquire data pertaining to traffic movement, accidents, snarl-ups, and climatic disruptions like landslides and waterlogging. All this keeps the drivers informed in real time, ensuring their safety, and at the backend allows better understanding and management of traffic. Not just that, the ITS deployment can also facilitate safe and efficient freight transportation, thus improving the performance of supply chains.

According to MarkNtel Advisors, a research, consulting and data analytics firm, India's ITS market is projected to grow at a CAGR, or compound annual growth rate, of around 16 per cent between 2022 and 2027. "The market is driven by the swift improvement in the economic conditions of the people and their inclination toward private ownership of vehicles... becoming a prominent factor behind the rising number of road accidents. It, in turn, is augmenting the demand for ITS to minimise such possibilities," it notes in a July 2022 report.

In the past few years, India has added about 9,000 km of highways loaded with these gizmos. The aim is to reach 10,000 km by the end of FY24. And even state governments are working in this direction. In Tamil Nadu, for instance, the highways department is turning seven arterial roads of Chennai into smart roads with WiFi poles, CCTV cameras, information kiosks, etc. The aim is to reduce accidents and congestion while making the roads safer for non-motorised transport.

THE CHANGE-MAKERS

NHAI

The National Highway Authority of India is executing greenfield projects worth Rs 7 lakh crore, with smart technologies being deployed on these highways for incident management, emergency response, electronic toll payment, etc.

10,000 KM

Length of highways the GoI plans to cover with Intelligent Transport System by the end of FY24

STATE GOVERNMENTS

From Ujjain in Madhya Pradesh to Chennai in Tamil Nadu, state governments are commissioning smart roads for better management of traffic and cutting down on pollution

WHAT INDIA NEEDS TO MASTER

At about 145,000 km, India has the second largest network of national highways in the world, after the US. And this makes it important for the government to expand its ITS network too. But to be widely adopted, these smart solutions must be cost-effective, easy to implement, and need minimal human intervention.

Cost escalation, however, has remained the bane of many a project in India. Also, the heaps of varied data collected is not always integrated (language fluctuations, to name one cause) and require a skilled workforce to analyse it and make timely interventions. Other challenges include dealing with data privacy or convincing people to accept the new technology. Like FASTag, first rolled out for automatic toll fee collection in 2016, was finally made mandatory only in 2021. It may be gradual, but there's progress for sure. ■